STEVEN YANG

(647) 469-6927 | stevenyang.142857@gmail.com | Website | LinkedIn

Education

University of Toronto **Sep. 2024 – May 2028 (Expected)**

- *Honours Bachelor of Science*, Data Science Specialist, Computer Science Major
- cGPA: 4.0/4.0
- **Awards:** Dean's List Scholar (2025, 2026) and Woodsworth College Scholarship Fund (2026)
- Relevant coursework: Software Design, Software Tools and System Programming, Data Structures and Analysis, Data Science, Probability and Statistics, Linear Algebra, Multivariable Calculus

Research

Student Researcher **Sep. 2026 – Present**

University of Toronto

- Build **AgentLangBench** to evaluate code generation, formal verification, and runtime performance across programming languages for AI agents.
- Surveyed 42 languages and tools for AI agents. Used pinned toolchains and repeatable Linux tests to identify 16 entries that passed all applicable checks without workarounds.
- Audited AI-generated solutions across languages to assess whether their contracts captured the original task requirements.

Experience

Research and Development Intern **May 2026 – Aug. 2026**

NSFOCUS *Beijing, China*

- Built a Python pipeline using the **Tavily** and **FOFA** APIs to find suspicious domains and social media accounts, remove duplicates, and score each result by risk.
- Prototyped a **DeepSeek** workflow to match threat events to client profiles, remove duplicates, and determine which events to deliver.
- Wrote a research report analyzing the architecture and workflow of an AI coding agent.

Projects

NHANES Supplement Use Analysis *Python, pandas, NumPy, matplotlib, scikit-learn* **2026**

- Worked with a teammate to analyze how supplement use varied with diet and demographic characteristics in the 2021–2023 NHANES survey.
- Cleaned and preprocessed linked survey datasets and engineered features for modeling.
- Modeled supplement use with statistical and **machine learning** methods and visualized predictive factors and model performance.

Question Bank Platform *Java, OpenAI API* **2026**

- Built a full-stack Java application to extract and structure math exam questions from PDFs into a searchable digital database.
- Combined **OCR** and OpenAI vision models to extract question text, diagrams, and metadata from page images.
- Designed an asynchronous pipeline to segment PDFs, store results as JSON, and flag low-confidence extractions for review.

GroupFlow *Java, Swing, MongoDB, JUnit* **2025**

- Designed and implemented a collaborative task and group management application using **Clean Architecture**, separating entities, use cases, and UI layers.
- Built core features including group creation, role-based permissions, task assignment, and progress tracking.
- Used interfaces to isolate components for unit testing with **JUnit**.
- Reviewed teammates' code and integrated changes through Git pull requests.

Technical Skills

- **Languages:** Python, Java, C, JavaScript, Rust, HTML/CSS
- **Developer Tools:** Codex, Claude Code, Git/GitHub, Linux, VS Code, IntelliJ, Jupyter, LaTeX, PostgreSQL, Docker, Verus, Dafny
- **Frameworks/Libraries:** NumPy, pandas, matplotlib, scikit-learn, SciPy, statsmodels, TensorFlow